

Solve statistical problems

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Dataset A has mean 24 over 8 values. Dataset B has total 207 over 9 values. Which mean is greater? Copy or complete the first step.

2. A line graph rises from 18 to 46. What is the increase?

3. A quarter of a 240-pupil pie chart represents walkers. How many pupils walk?

4. Miro says: Miro divides by the largest value instead of the number of values when finding the mean. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Dataset A has mean 24 over 8 values. Dataset B has total 207 over 9 values. Which mean is greater? Copy or complete the first step.

Answer Dataset A: 24; Dataset B: 23, so A is greater.

2. A line graph rises from 18 to 46. What is the increase?

Answer 28.

3. A quarter of a 240-pupil pie chart represents walkers. How many pupils walk?

Answer 60 pupils.

4. Miro says: Miro divides by the largest value instead of the number of values when finding the mean. Is this correct? Explain.

Answer Add every value, count how many values there are, then divide the total by that count.

Solve statistical problems

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Dataset A has mean 24 over 8 values. Dataset B has total 207 over 9 values. Which mean is greater?

6. Check this result using an inverse, estimate or known fact: A quarter of a 240-pupil pie chart represents walkers. How many pupils walk?

2. A line graph rises from 18 to 46. What is the increase?

7. Prove or explain your reasoning: In a survey of 60 pupils, 15 choose art. Find the pie-chart angle.

3. A quarter of a 240-pupil pie chart represents walkers. How many pupils walk?

8. Identify and correct this misconception: Miro divides by the largest value instead of the number of values when finding the mean.

4. Solve this independently and show every stage: Dataset A has mean 24 over 8 values. Dataset B has total 207 over 9 values. Which mean is greater?

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: A line graph rises from 18 to 46. What is the increase?

10. Write and solve a similar question to: A quarter of a 240-pupil pie chart represents walkers. How many pupils walk?

Teacher answers and checking prompts

1. Dataset A has mean 24 over 8 values. Dataset B has total 207 over 9 values. Which mean is greater?
Answer Dataset A: 24; Dataset B: 23, so A is greater.
2. A line graph rises from 18 to 46. What is the increase?
Answer 28.
3. A quarter of a 240-pupil pie chart represents walkers. How many pupils walk?
Answer 60 pupils.
4. Solve this independently and show every stage: Dataset A has mean 24 over 8 values. Dataset B has total 207 over 9 values. Which mean is greater?
Answer Dataset A: 24; Dataset B: 23, so A is greater.
5. Use a second method or representation for: A line graph rises from 18 to 46. What is the increase?
Answer Expected result: 28. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: A quarter of a 240-pupil pie chart represents walkers. How many pupils walk?
Answer Expected result: 60 pupils. A valid check must be shown.
7. Prove or explain your reasoning: In a survey of 60 pupils, 15 choose art. Find the pie-chart angle.
Answer 90 degrees. The explanation must show why the method works.
8. Identify and correct this misconception: Miro divides by the largest value instead of the number of values when finding the mean.
Answer Add every value, count how many values there are, then divide the total by that count.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: A quarter of a 240-pupil pie chart represents walkers. How many pupils walk?
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Solve statistical problems

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: In a survey of 60 pupils, 15 choose art. Find the pie-chart angle.

6. Create a new example that tests the same idea as: A quarter of a 240-pupil pie chart represents walkers. How many pupils walk?

2. Prove the answer to this question, rather than only calculating it: Dataset A has mean 24 over 8 values. Dataset B has total 207 over 9 values. Which mean is greater?

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: A line graph rises from 18 to 46. What is the increase?

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 60 pupils.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro divides by the largest value instead of the number of values when finding the mean.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: In a survey of 60 pupils, 15 choose art. Find the pie-chart angle.
Answer 90 degrees. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Dataset A has mean 24 over 8 values. Dataset B has total 207 over 9 values. Which mean is greater?
Answer Expected result: Dataset A: 24; Dataset B: 23, so A is greater. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: A line graph rises from 18 to 46. What is the increase?
Answer Expected result: 28. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 60 pupils.
Answer One valid reconstruction is: A quarter of a 240-pupil pie chart represents walkers. How many pupils walk? Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro divides by the largest value instead of the number of values when finding the mean.
Answer Add every value, count how many values there are, then divide the total by that count.
6. Create a new example that tests the same idea as: A quarter of a 240-pupil pie chart represents walkers. How many pupils walk?
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.